

STA 640 — Causal Inference

Chapter 4.2 Treatment Effect Heterogeneity: Machine Learning Approaches

Fan Li

Department of Statistical Science
Duke University

CATE estimation: basic interaction approaches

- ▶ Subgroup analysis: subgroups are **pre-specified**, static
- ▶ The literature has increasingly moved towards identify subgroups with significant effects **post-analysis**, dynamic
- ▶ As discussed earlier, under unconfoundedness

$$\mathbb{E}[Y(1) - Y(0)|X = x] = \mathbb{E}[Y|Z = 1, X = x] - \mathbb{E}[Y|Z = 0, X = x]$$

- ▶ This implies we can simply build an outcome model for
$$f(z, x) = \mathbb{E}[Y|Z = z, X = x]$$
- ▶ Once we have estimates of this outcome model, we have estimates of the CATE $\hat{\tau}(x) = \hat{f}(1, x) - \hat{f}(0, x)$

CATE estimation: Basic interaction approaches

- ▶ In principle, any outcome regression model (e.g. a simple linear regression) can be used to calculate CATE
- ▶ The simplest approach is with a linear model

$$f(Z, X) = \beta_0 + \beta_x X + \beta_z Z + \beta_{zx} ZX$$

- ▶ Related approaches for other models, such as SVMs (Imai and Ratkovic, 2013)
- ▶ Easy to see that $\tau(x) = \beta_z + \beta_{zx} X$
- ▶ If we center X , then the ATE is simply

$$E(Y(1) - Y(0)) = \beta_z$$

- ▶ Otherwise the ATE is given by

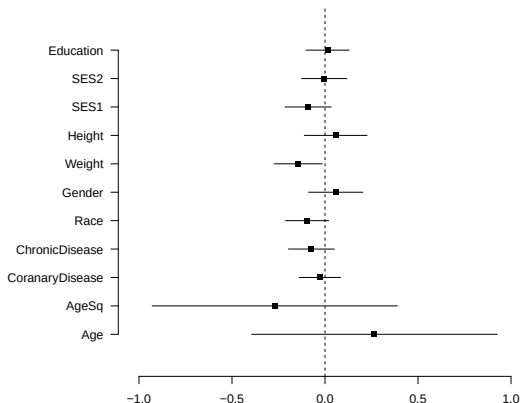
$$E(Y(1) - Y(0)) = \beta_z + \beta_{zx} \mathbb{E}(X)$$

CATE estimation: Basic interaction approaches

- ▶ Two main reasons why one might like this approach
 - ▶ Simple and easy to implement
 - ▶ Very interpretable
- ▶ A lot of questions are easy to answer in this framework
- ▶ Which covariates modify the treatment effect most
 - ▶ Examine magnitude of individual β_{zx} values
- ▶ Is there any treatment effect heterogeneity?
 - ▶ Amounts to testing $H_0 : \beta_{zx} = 0$

CATE estimation: Basic interaction approaches

- ▶ Below are estimates of β_{zx} from the NHANES analysis
- ▶ Overall ATE is estimated to be -0.08 (-0.19, 0.03)
 - ▶ More pronounced, negative effect in individuals with higher weight



CATE estimation: Basic interaction approaches

- ▶ A very related approach is to specify separate models in the treated and control groups

$$f(1, X) = \beta_{01} + \beta_{x1}X$$

$$f(0, X) = \beta_{00} + \beta_{x0}X$$

- ▶ The CATE is therefore

$$\tau(x) = \beta_{01} - \beta_{00} + (\beta_{x1} - \beta_{x0})x$$

- ▶ Treated individuals used to estimate $f(1, X)$ and vice-versa
- ▶ In linear models, these two approaches are identical
- ▶ Once we jump to nonlinear, flexible approaches these two will behave much differently

Flexible CATE estimators

- ▶ There has been a dramatic increase in semiparametric or nonparametric estimators of the CATE that utilize modern statistical learning tools
 - ▶ Bayesian nonparametric approaches
 - ▶ Machine learning (trees, regularized regressions, boosting, etc.)
- ▶ Throughout the rest of the lecture, we will review many of these approaches
 - ▶ Discuss pros and cons of each
- ▶ Some are left out, but this will cover many of the core ideas

High-dimensional causal analysis and machine learning

- ▶ High dimensional settings is common in CATE estimation, but also in ATE estimation
- ▶ Two types of high-dimensional settings:
 - ▶ A large number of covariates
 - ▶ A (propensity or/and outcome) model with a large number of parameters, regardless of the number of covariates, e.g. nonparametric and semiparametric models
- ▶ In both cases, machine learning (ML) are often used: dimension reduction, regularized inference
- ▶ ML methods are designed for prediction, how about causal, i.e. counterfactual prediction, task? Turns out to be not straightforward

Double/debiased machine learning

- ▶ A recurring idea in causal inference with high dimensional data or machine learning is “double” learning: using ML for both outcome and propensity model, and combine
- ▶ A general theoretical framework is double/debiased ML by Chernozhukov et al.
- ▶ Recall: ML methods are effective for prediction, how about causal (i.e. counterfactual prediction) task?
- ▶ Good prediction performance of ML models does not automatically translate into good performance for estimation of “causal” parameters
 - ▶ Regularization bias: slower convergence rate
 - ▶ Overfitting bias: Capturing more than the relationship between Y and Z

An Earlier Example: Double Selection for Causal Inference

- ▶ An earlier method is to use ML methods in double-robust (DR) estimators for ATE
- ▶ **Main idea:** specify ML models for both propensity score and outcome models (Farrell, 2015)
- ▶ With high-dimensional confounders, Belloni et al. (2014) proposes a double-selection procedure
 - ▶ Select confounders/covariates for the propensity score model and for the outcome model, e.g., by LASSO
 - ▶ Use least square estimation of the outcome with treatment indicator plus **the union of selected confounders**
- ▶ “Double-selection” gives $\sqrt{N}$ consistency of ATE, whereas “single-selection” cannot reach

Frisch-Waugh-Lovell (FWL) Theorem

- ▶ Consider a linear regression

$$Y = \beta_0 + \beta_1 Z + \beta_2 X + U,$$

with $E(U|Z, X) = 0$

- ▶ To obtain β_1 , we can use OLS by concatenating Z, X
- ▶ Frisch-Waugh-Lovell theorem gives another consistent way to estimate β_1 (partial regression):
 1. Regress (linear) Y on X , obtain residual $\hat{U} = Y - \hat{Y}$
 2. Regress (linear) Z on X , obtain residual $\hat{V} = Z - \hat{Z}$
 3. Regress (linear) $\hat{U}$ on $\hat{V}$, obtain $\hat{\beta}_1$
- ▶ Classic result in linear models, see Chap 7 in Ding (2025) **here (click)**

Robinson decomposition

- ▶ Robinson (1988) generalized FWL theorem: replace the linear regressions in FWL to some nonparametric (e.g. kernel) regression
- ▶ Robinson's procedure:
 1. Kernel regression of Y on X , obtain residual $\hat{U} = Y - \hat{Y}$
 2. Kernel regression of Z on X , obtain residual $\hat{V} = Z - \hat{Z}$
 3. Regress (linear) $\hat{U}$ on $\hat{V}$, obtain $\hat{\beta}_1$
- ▶ Double machine learning (DML) further generalizes FWL and Robinson ideas to machine learning models

Canonical Example of DML: Partial Linear Model

Chernozhukov et al. 2018

- ▶ Intuition: The relationship between Y (outcome) and X (covariates) is usually more complex than the relationship between Y and Z (treatment)
- ▶ Idea: Use a ML model for $Y \sim X$ and a linear model for $Y \sim Z$
- ▶ Consider a causal partial linear model:

$$Y = Z\tau + \mu(X) + U$$

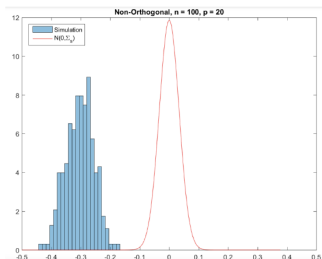
$$Z = e(X) + V$$

with $\mathbb{E}(U|X, Z) = 0$ and $\mathbb{E}(V|X) = 0$

- ▶ X : a high-dimensional vector of covariates/confounders
 $\mathbb{E}(U|X, Z) = 0$
- ▶ τ : causal estimand ATE; later τ can be extended to CATE, $\tau(X)$
- ▶ $\mu(X), e(X)$: nuisance (and often complex) parameters

Näive prediction-based ML approach is Bad

- ▶ Predict Y using X and Z : $\hat{Y} = Z\hat{\tau} + \hat{\mu}(X)$
- ▶ We can estimate τ and $\mu(x)$ iteratively:
 - ▶ Given initial parameters, (i) run a ML model, e.g. random forest or boosting on $\hat{Y} - Z\hat{\tau}$ to fit $\hat{\mu}(X)$; (ii) run OLS on $\hat{Y} - \hat{\mu}(X)$ to fit $\hat{\tau}$
 - ▶ Repeat until converge
- ▶ Good prediction performance of Y : small $(\hat{Y} - Y)^2$, but dist of the estimated causal parameter, $\hat{\tau} - \tau$, looks like this



Source of bias and solutions

- ▶ We can look at the asymptotic distribution of $\hat{\tau} - \tau$; a clear derivation is [here \(click\)](#)
- ▶ Two sources of bias
 - ▶ Regularization bias: Machine learning methods employ regularization (e.g., L1 or L2 regularization) to reduce variance, but this often induces bias and slower convergence rates
 - ▶ How to solve? Double ML - using ML twice: once to learn Y on X , and once to learn Z on X , and then regress residual on residual – the FWL/Robinson style
 - ▶ Overfitting bias
 - ▶ How to solve? Sample-splitting and cross-fitting.
- ▶ The key theory were developed in a series of papers by Chernouzhukov and co-authors, starting from Chernouzhukov et al. (2017, 2018)

DML Algorithm - Summary

In summary, for a given dataset $\{X_i, Z_i, Y_i\}_{i=1}^N$, DML follows this algorithm to estimate average treatment effect:

- 1 *Split sample*: random partition the data into k mutually exclusive parts: $\{I_k\}_{k=1}^K$. For each k , define $I_k^c = \{1, \dots, N\} \setminus I_k$
- 2 *Estimate propensity model in training sample*: Train any (regularized) ML model M_z to predict Z from X (propensity) using auxiliary I_k^c
- 3 *Estimate outcome model in training sample*: Train any (regularized) ML model M_y to predict Y from X (outcome) using auxiliary I_k^c
- 4 *Estimate ATE in prediction sample*: obtain the residuals in I_k : $Z_R = Z - M_z(X)$, and $Y_R = Y - M_y(X)$, and regress (linearly) Y_R on Z_R to estimate ATE
- 5 *Aggregate over K folds*: Repeat 2-3 for $k = 2, \dots, K$ so that DML uses the full data

DML for CATE

- ▶ Extend DML to CATE with a generalized partial linear model:

$$Y = Z\tau(X) + \mu(X) + U, \quad \mathbb{E}(U|X, Z) = 0$$

$$Z = e(X) + V, \quad \mathbb{E}(V|X) = 0$$

with $\mathbb{E}(UV|X, Z) = 0$, where $\tau(X)$ is the CATE

- ▶ Same idea: regress residualized outcome $\hat{U}$ and treatment $\hat{V}$:

$$\hat{\tau}(X) = \operatorname{argmin}_{\tau \in E_n} [(\hat{U} - \tau(X)\hat{V})^2]$$

- ▶ Difference choices of $\tau(X)$ in DML for CATE:
 - ▶ Reproducing Kernel Hilbert Space (Nie and Wager, 2021)
 - ▶ Random forest (Athey et al. 2019)
 - ▶ Sparse linear space (Chernozhukov et al.)
- ▶ A rich online source and package of DML is **here (click)**

R-learners

- ▶ R-learners (Nie and Wager, 2021): independently developed from DML; essentially a special case of DML for CATE
- ▶ Assuming the same generalized partial linear model (varying coefficient model)

$$Y_i = \tau(X_i)Z_i + \mu(X_i) + \epsilon_i$$

and if we take the conditional expectation of this, we obtain

$$m(X_i) = E(Y_i \mid X_i) = \mu(X_i) + \tau(X_i)e(X_i)$$

- ▶ Subtracting the two equations cancels out the (complex) nuisance function $\mu(x)$ (the insight first pointed out in Robinson (1988))

$$Y_i - m(X_i) = (Z_i - e(X_i))\tau(X_i) + \epsilon_i$$

R-learners

- ▶ Which implies that

$$\tau(\cdot) = \underset{\tau}{\operatorname{argmin}} \left\{ \mathbb{E} \left[\left((Y_i - m(X_i)) - (Z_i - e(X_i))\tau(X_i) \right)^2 \right] \right\}$$

- ▶ Nie and Wager (2021)'s main idea: estimate the CATE in the following way:

$$\tau(\cdot) = \underset{\tau}{\operatorname{argmin}} \left\{ \widehat{L}_n(\tau(\cdot)) + \Lambda_n(\tau(\cdot)) \right\}$$

where

$$\widehat{L}_n(\tau(\cdot)) = \frac{1}{n} \sum_{i=1}^n \left((Y_i - \widehat{m}^{-i}(X_i)) - (Z_i - \widehat{e}^{-i}(X_i))\tau(X_i) \right)^2$$

- ▶ $\Lambda_n(\tau(\cdot))$ is a penalty on the complexity of the CATE
 - ▶ Many options such as smoothness penalties, lasso, etc.

R-learners

- ▶ Note that we used $\widehat{m}^{-i}(X_i)$ and $\widehat{e}^{-i}(X_i)$ in the squared error loss
- ▶ These are estimates of the conditional mean outcome regression and propensity score with the i^{th} observation removed
 - ▶ Typically done using 5 or 10-fold cross validation, not leave one out
- ▶ This approach separated the problem into two separate stages
 - ▶ Estimating nuisance functions, $m(\cdot)$ and $e(\cdot)$
 - ▶ Estimation of $\tau(\cdot)$ conditional on nuisance function estimates
- ▶ Allows for separate penalization in these two steps
 - ▶ Allows for the CATE to be much simpler than the outcome regression functions

R-learners

- ▶ Key components:
 - ▶ Regress residuals of outcome on residuals of treatment – special case of DML
 - ▶ Varying coefficient model, different complexity for baseline counterfactual $\mu(x)$ (equivalently $Y(0)$) and treatment effect $\tau(x)$
- ▶ Nie and Wager (2021) show R-learners can be used with many modern machine learning type of estimators for the CATE
 - ▶ High-dimensional models
 - ▶ Gradient boosting
 - ▶ Neural networks

Another double learning approach: TMLE

van der Laan and Rubin (2006) and series of work

► Targeted Maximum Likelihood Estimation (TMLE)

1. Initial outcome model: estimate $\{\hat{f}^{(0)}(Z, X)\}$ using your preference ML algorithm; generate initial prediction: $\{\hat{f}^{(0)}(1, X_i)\}$ and $\{\hat{f}^{(0)}(0, X_i)\}$ for each unit i
2. Fit a parametric (or ML) PS model $\hat{e}(x)$; compute the “**clever covariate**” (IPW): $W_i(Z, X) = Z_i/\hat{e}(X_i) + (1 - Z_i)/(1 - \hat{e}(X_i))$
3. Targeting/fluctuation step: Fit a canonical GLM with link function $h(\cdot)$ of Y_i on W_i with an offset term $h(\hat{f}^{(0)}(Z_i, X_i))$: **the estimated coefficient $\hat{\epsilon}$ of W_i captures the residual bias in the initial outcome model**
4. Update the outcome model:

$$\hat{f}^*(Z, X) = h^{-1}\{h(\hat{f}^{(0)}(Z, X))\} + \hat{\epsilon} \cdot W(Z, X)$$

5. Plug-in estimator for the ATE: $\hat{\tau} = \sum_i [\hat{f}^*(1, X_i) - \hat{f}^*(0, X_i)]/n$

Another double learning method: TMLE

- ▶ The GLM in step (3) is called a **fluctuation working model**
- ▶ Without the fluctuation model, the algorithm is simply an OR estimator based on $N^{-1} \sum_{i=1}^N \hat{f}^{(0)}(X_i)$
- ▶ TMLE uses (inverse of) PS to fluctuate the initial regression
 - ▶ can show that the score of the stabilized fluctuation model at zero fluctuation ($\hat{\epsilon}_n = 0$) spans the doubly robust estimating function (recall discussion earlier on regression with the **clever covariate**)
- ▶ TMLE is a **fully iterated** double-robust estimator; compatible with ML models, can extend to CATE
- ▶ TMLE: inference via efficient influence curve, reaching the Cramér-Rao efficiency bound for semiparametric models

CATE estimation: S-Learners

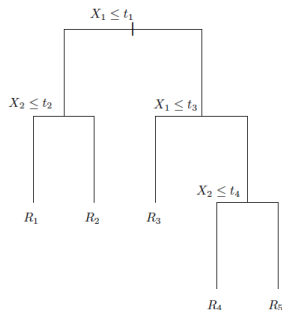
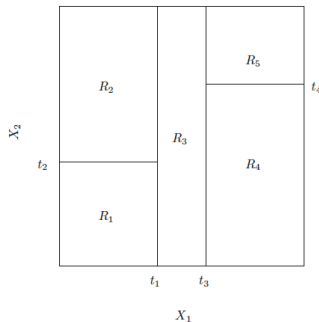
- ▶ One class of outcome modeling approach is sometimes referred to as S-learners (S refers to single)
- ▶ Exploit the fact that (recall notation $f(z, x) = \mathbb{E}[Y|Z = z, X = x]$)

$$\tau(x) = f(1, x) - f(0, x)$$

- ▶ Focus solely on flexible estimation of $f(z, x)$
 - ▶ CATE estimation is automatic after this
- ▶ Countless ML approaches to estimating $f(z, x)$; trees and forests are popular, both frequentists (e.g. CART, random forest) and Bayesian (e.g. BART)

Brief review of regression trees

- ▶ Regression trees partition the covariate space into non-overlapping regions
- ▶ Predictions in each region based solely on data that fall in that region, R_j



James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). An introduction to statistical learning. New York: springer.

BART for CATE estimation

- ▶ Hill (2011): first paper to use BART to estimate $f(z, x)$
- ▶ BART (Chipman et al. 2010): essentially an ensemble of trees with the partition and depth regularized (shrunk) by a prior

$$f(z, x) = \sum_{t=1}^T g(x, z; \mathcal{T}_t, \mathcal{M}_t)$$

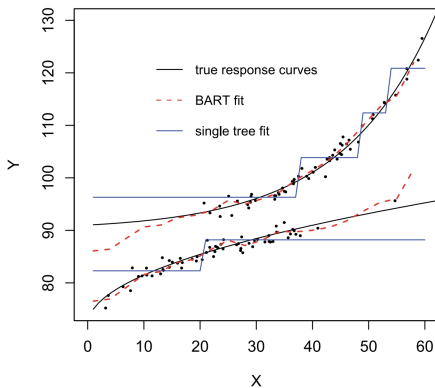
- ▶ Here, $g(x, z; \mathcal{T}_t, \mathcal{M}_t)$ is a tree that partitions the space of x and z
 - ▶ $\mathcal{T}_t$: represents the tree structure (where splits are)
 - ▶ $\mathcal{M}_t$: parameters for predictions in each terminal node of the tree
- ▶ $\mathcal{M}_t = (\mu_{t1}, \dots, \mu_{tL_t})$ where L_t is the number of terminal nodes

BART for CATE estimation

- ▶ BART is a Bayesian approach, placing priors on the parameters of the trees to regularize
- ▶ The prior probability of splitting decreases with tree depth
 - ▶ Probability of splitting at node depth k is $\gamma(1+k)^{-\beta}$ with $\gamma, \beta > 0$
- ▶ Shrinkage of mean parameters in each terminal node are shrunk by a factor of T
 - ▶ $\mu_{tl} \sim \mathcal{N}(0, \sigma_u^2/T)$
- ▶ Empirical evidence: BART outperforms random forests (Chipman et al., 2010)
 - ▶ Inference also easy in the Bayesian paradigm
 - ▶ Effectively tuning parameter free (defaults work well)
 - ▶ Cons: computationally intensive, as all Bayesian methods

BART for CATE estimation

- ▶ BART is an ensemble of trees, essentially a Bayesian version of random effects with shrinkage on tree structure: much better than using a single regression tree
 - ▶ Not surprising given performance of boosting or RFs compared to a single tree



Bayesian Causal Forest (Hahn et al., 2020)

- ▶ Drawbacks for putting a BART prior on the response function $f(z, x)$: not differentiate between nuisance parameters and the parameter of interest, $\tau(x)$
 - ▶ Do we expect the CATE $\tau(x)$ to be as complex as the baseline response $f(0, x)$?
- ▶ Bayesian Causal Forest (BCF): use the same partial linear model in DML and R-learners (independently developed by Hahn et al.)

$$f(z, x) = \mu(x) + \tau(x)z$$

where **separate BART priors** placed on $\mu(x)$ and $\tau(x)$

- ▶ can use simpler trees for $\tau(x)$

Bayesian Causal Forest (Hahn et al., 2020)

- ▶ Hahn et al. advocate for including the **propensity score** in $\mu(x)$

$$f(z, x) = \mu(x, \widehat{e}(x)) + \tau(x)z$$

- ▶ Give simulation and empirical evidence: adding PS $\widehat{e}(x)$ mitigate an issue called regularization induced confounding (bias)
 - ▶ Unintended bias that occurs when we are not careful about how we implement regularization or shrinkage in high-dimensional or nonparametric situations
 - ▶ Our model might indirectly shrink degree of confounding bias to zero, which is bad when there is severe confounding

Choice of priors

- ▶ BART is a special case of the Bayesian nonparametric model.
There are others, e.g. Gaussian Process (GP), Dirichlet Process mixture (DPM)
- ▶ Which one to choose? Depends on the degree of overlap
- ▶ A desirable prior should accurately reflect uncertainty for various degree of overlap
- ▶ Simulation evidence:
 - ▶ In regions with good overlap, all methods perform similarly
 - ▶ In regions with poor overlap, choose a robust prior that adaptively captures uncertainty according to different degree of overlap.
 - ▶ With poor overlap, BART appears to struggle whereas GP and DPM do better in reflecting uncertainty

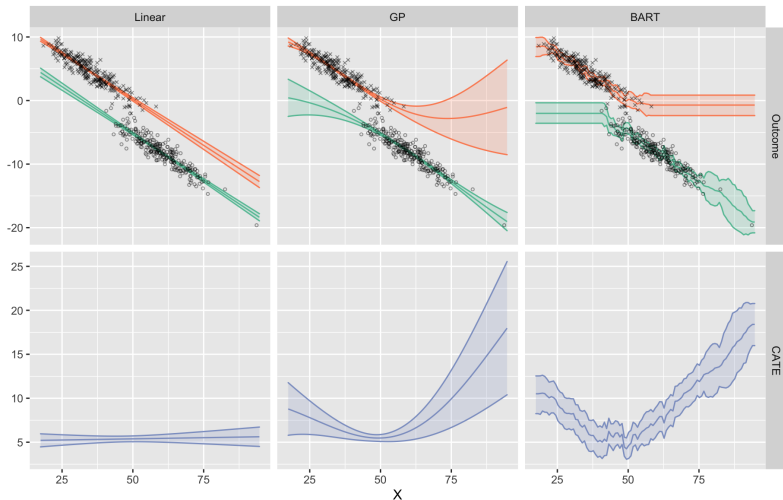
Choice of Priors: a simulated example

- ▶ (An example first due to Surya Tokdar, details in Li et al. (2022) review paper)
- ▶ A study with 250 treated and 250 control units
- ▶ A single covariate X following Gamma distribution with mean 60 in the control and 35 in the treatment group, and with SD 8 in both groups.
- ▶ A true outcome model with constant treatment effects:

$$Y_i(z) = 10 + 5z - 0.3X_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1)$$

- ▶ here the CATE $\tau(x) = 5$ for all x . Covariate overlap is good between the groups in the middle of the range of X (around 40 to 50), but deteriorates towards the tails of X .

Choice of Priors: a simulated example



T-learner

- ▶ A natural extension of the S-learner is the T-learner (T refers to “two”)
- ▶ The previous approach used all of the data to fit one model

$$E(Y \mid Z = z, X = x) = f(z, x)$$

- ▶ A T-learner fits separate models to the treated and control groups

$$E(Y \mid Z = 1, X = x) = f_1(x)$$

$$E(Y \mid Z = 0, X = x) = f_0(x)$$

and the CATE is simply

$$\tau(x) = f_1(x) - f_0(x)$$

T-learner

- ▶ A couple advantages to this approach
 - ▶ Extremely flexible
 - ▶ Works well when $f_z(x)$ differs greatly across $z = 0, 1$
- ▶ Some drawbacks as well
 - ▶ Too flexible! Highly variable
 - ▶ Difficult to estimate $f_z(x)$ when treatment group z has few individuals
 - ▶ Again no control of $\tau(x)$

T-learner

- ▶ Suppose we estimate $f_z(x)$ separately in each group, we have

$$\text{Var}(\widehat{f}_1(x)) = v_1, \quad \text{Var}(\widehat{f}_0(x)) = v_0$$

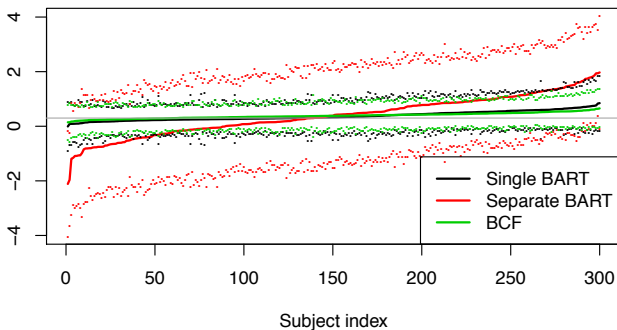
- ▶ Due to independence of individuals

$$\text{Var}(\widehat{\tau}(x)) = v_0 + v_1$$

- ▶ The variance of the treatment effect is greater than both of the individual functions!
 - ▶ Does this coincide with our prior knowledge about the treatment effect function?
 - ▶ We generally expect the treatment effect to be as simple, or simpler than $f_z(x)$

T-learner

- ▶ Below are estimates and confidence intervals for $\tau(X_i)$ for $i = 1, \dots, n$ in a simulated data set with no heterogeneity
- ▶ Separate BART models leads to extremely wide intervals and variable estimates



T-learner

- ▶ There are many ways to address this problem
- ▶ One way is to impose some structure on $f_z(x)$
 - ▶ Put shrinkage directly on $\tau(x)$ as in Hahn et al. (2020)
 - ▶ R-learners, which use a specific loss function and a penalty on $\tau(x)$
 - ▶ Multi-task learners put shared structure on $f_1(x)$ and $f_0(x)$, e.g. a Gaussian Process (Alaa et al. 2017)
- ▶ Another line of approaches constructs pseudo-outcomes and regresses them against X
 - ▶ Connections to IPW and DR estimators
- ▶ Some approaches directly estimate the CATE
 - ▶ Causal forests, related tree-based approaches

Pseudo-outcomes

- ▶ The R-learner can be thought of as running a regression on a transformed outcome and covariate
 - ▶ Residualized outcome and treatment
 - ▶ Allow for coefficient in front of treatment to vary by X_i
- ▶ There are a number of other approaches that fit into the scope of transformed outcome regression
- ▶ These approaches run a regression on X , but use a special outcome that allows us to estimate $\tau(x)$
- ▶ Close connection with IPW and DR estimators from earlier lectures

Pseudo-outcomes

- ▶ Remember for estimating the ATE, we had that

$$\mathbb{E} \left[\frac{ZY}{e(X)} - \frac{(1-Z)Y}{1-e(X)} \right] = \tau^{\text{ATE}}.$$

- ▶ This motivated the IPW estimator, which is a sample average of this quantity

$$\hat{\tau}_{ipw} = \frac{1}{N} \left\{ \sum_{i=1}^N \frac{Y_i Z_i}{e(X_i)} - \sum_{i=1}^N \frac{Y_i (1 - Z_i)}{1 - e(X_i)} \right\}$$

Pseudo-outcomes

- ▶ It turns out that

$$\mathbb{E} \left[\frac{ZY}{e(X)} - \frac{(1-Z)Y}{1-e(X)} \middle| X = x \right] = \tau(x)$$

- ▶ We can define the transformed outcome as

$$O_i = \frac{Y_i Z_i}{e(X_i)} - \frac{Y_i (1 - Z_i)}{1 - e(X_i)}$$

- ▶ Essentially, O_i is an unbiased estimator of an individual's treatment effect
 - ▶ Hence $E(O_i) = \tau^{ATE}$ and $E(O_i \mid X_i = x) = \tau(x)$

Pseudo-outcomes

- ▶ IPW is not the only choice, we can also use the DR transformed outcome

$$O_i = \left\{ \frac{Z_i Y_i}{e(X_i)} - \frac{Z_i - e(X_i)}{e(X_i)} m_1(X_i) \right\} - \left\{ \frac{(1 - Z_i) Y_i}{1 - e(X_i)} + \frac{Z_i - e(X_i)}{1 - e(X_i)} m_0(X_i) \right\}$$

- ▶ If either $e(X_i) = P(Z = 1|X = X_i)$ or $m_z(X_i) = E(Y|Z = z, X = X_i)$ then

$$E(O_i|X = x) = \tau(x)$$

- ▶ This gives us doubly robust estimators of the CATE!
 - ▶ Assuming that we correctly specify the $\tau(\cdot)$ function as well

Pseudo-outcomes

- ▶ As with the R-learner, this lends itself to a two-stage estimation strategy
- ▶ First we need to construct estimates of the PS and outcome regression functions
 - ▶ Same as for ATE estimation
 - ▶ Use these to create O_i
- ▶ Then, once we have O_i , we can run a standard regression of O_i against X_i
 - ▶ Using any technique you want!
 - ▶ Inference is easier if you use a parametric model here

Pseudo-outcomes

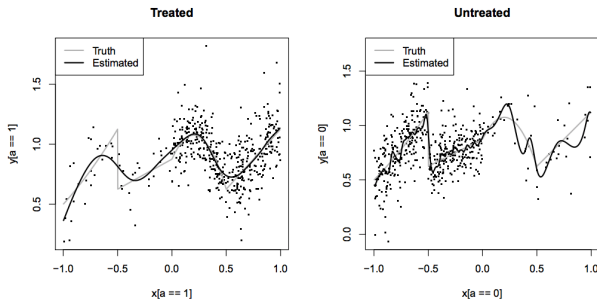
- ▶ Many nice features of this approach
- ▶ Solves the problem of the T-learner that the CATE is too complicated
 - ▶ Second stage estimates can be as simple as you want, regardless of the complexity of $e(X)$ or $m_z(X)$
- ▶ Doubly robust version allows us to estimate $\tau(x)$ at a faster convergence rate than either of $e(X)$ or $m_z(X)$ if they're both correctly specified (later)
- ▶ We can estimate a wide variety of estimands, not just $\tau(x)$

Pseudo-outcomes

- ▶ Not always interested in $E(Y(1) - Y(0)|X = x)$
- ▶ What if we only care about heterogeneity by a particular covariate, X_j ?
 - ▶ Very interpretable estimand that is relevant in many studies
- ▶ The pseudo-outcome framework accommodates this easily
 - ▶ The second stage regression can simply be a univariate one
- ▶ More generally we can learn heterogeneity by V instead of X , where V need not be a subset of X
 - ▶ Typically $V \subset X$

Pseudo-outcomes

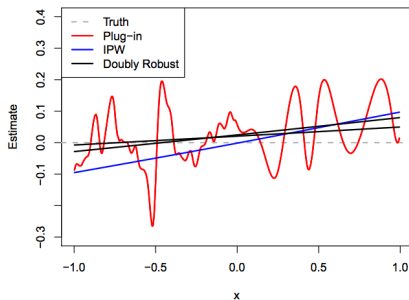
- ▶ Here is an illustration of where the individual regression functions are complex, but the difference between them is zero
- ▶ T-learner would simply take the difference between these two estimates



Kennedy, EH. "Towards optimal doubly robust estimation of heterogeneous causal effects." (2023)

Pseudo-outcomes

- ▶ Now here are the estimates of the CATE using the T-learner (Plug-in) and some pseudo-outcome approaches
- ▶ Pseudo-outcome approaches adapt to the simplicity of the problem much better



Kennedy, EH. "Towards optimal doubly robust estimation of heterogeneous causal effects." (2023)

Pseudo-outcomes

- ▶ Strong theoretical support
- ▶ If we define the oracle risk as

$$R^*(x) = \mathbb{E} \left[\{ \tilde{\tau}(x) - \tau(x) \}^2 \right]$$

with $\tilde{\tau}(x)$ comes from regressing the true $Y_i(1) - Y_i(0)$ against X_i

- ▶ Then under certain conditions, the pseudo-outcome with the DR construction satisfies

$$\mathbb{E} \left[\{ \hat{\tau}(x) - \tau(x) \}^2 \right] \leq R^*(x) + \mathbb{E} \left[\{ \hat{e}(x) - e(x) \}^2 \right] \sum_{z=0}^1 \mathbb{E} \left[\{ \hat{m}_z(x) - m_z(x) \}^2 \right]$$

- ▶ However, transformed outcome inherits the extreme weights problem of IPW and the empirical performance is often sensitive to lack of overlap and inferior to other methods

Causal trees and forests

- ▶ Tree-based (frequentist) approaches are popular; directly estimate the CATE
- ▶ The overarching goal of these approaches is to find subsets of the data where the **treatment effect varies the most**
- ▶ No need to specify functional form for $\tau(x)$
 - ▶ Assumed constant within areas of covariate space
- ▶ Key papers in this area are Wager and Athey (2018), Athey et al. (2019), and Powers et al. (2018)

A quick remark on regression trees for prediction

- ▶ How do we determine the tree structure in prediction literature?
 - ▶ Which covariates to split on?
 - ▶ What value of a covariate do we split at?
- ▶ Bayesian: add prior distribution to shrink estimates (see earlier)
- ▶ Frequentist: pick splits that minimize the **prediction** MSE the most among all possible splits
 - ▶ Or Gini index / classification error for categorical outcomes
- ▶ Greedy algorithm that successively creates splits that improve the model the most

A quick remark on regression trees for prediction

- ▶ Suppose I'm at the top of a tree and haven't split yet
- ▶ My current prediction is $\hat{Y}_i = \bar{Y}$ for all i
- ▶ Now we find the values of j and s that minimize

$$\sum_{i: X_i \in R_1(j, s)} (Y_i - \hat{Y}_{R_1})^2 + \sum_{i: X_i \in R_2(j, s)} (Y_i - \hat{Y}_{R_2})^2$$

where

$$R_1(j, s) = \{X | X_j < s\} \quad R_2(j, s) = \{X | X_j \geq s\}$$

- ▶ Predictions in these regions are just sample averages (within group)

$$\hat{Y}_{R_1} = \frac{\sum_{i=1}^n \mathbf{1}\{X_i \in R_1(j, s)\} Y_i}{\sum_{i=1}^n \mathbf{1}\{X_i \in R_1(j, s)\}} \quad \hat{Y}_{R_2} = \frac{\sum_{i=1}^n \mathbf{1}\{X_i \in R_2(j, s)\} Y_i}{\sum_{i=1}^n \mathbf{1}\{X_i \in R_2(j, s)\}}$$

Causal trees and forests

- ▶ Causal trees are constructed in a similar way
- ▶ Key difference: instead of splitting to reduce MSE the most, we split to **maximize heterogeneity of the treatment effect** (later)
 - ▶ This will lead us to finding areas of the covariate space with different treatment effects
- ▶ Another difference: in the terminal nodes
 - ▶ In regression trees, the estimates are sample averages of the outcome
 - ▶ In causal trees, the estimates are the treatment effects
- ▶ There are multiple causal tree algorithms, but we will mostly focus on the original one from Wager and Athey (2018)

Causal trees: estimate treatment effects in nodes

- ▶ Suppose we have a tree with terminal nodes or leaves given by $L_1(x), \dots, L_K(x)$

- ▶ In leaf k , we can estimate the treatment effect as

$$\frac{1}{|\{i : X_i \in L_k(x), Z_i = 1\}|} \sum_{i: X_i \in L_k(x), Z_i = 1} Y_i \\ - \frac{1}{|\{i : X_i \in L_k(x), Z_i = 0\}|} \sum_{i: X_i \in L_k(x), Z_i = 0} Y_i$$

- ▶ The hope is that within leaf k , individuals have similar covariate values and therefore the treatment is as if randomized
 - ▶ And therefore the difference in means estimator is unbiased

Causal forests: where to split?

- ▶ Now suppose that we're considering a split of a parent node into two separate nodes
- ▶ The estimated treatment effects in each new node are given by $\widehat{\tau}_l$ and $\widehat{\tau}_r$
- ▶ One approach to finding splits is to calculate **heterogeneity of the treatment effect**:

$$\frac{|\widehat{\tau}_l - \widehat{\tau}_r|}{\sqrt{\widehat{\text{Var}}(\widehat{\tau}_l) + \widehat{\text{Var}}(\widehat{\tau}_r)}}$$

and choose the split that maximizes this

- ▶ Other approaches explored in Athey and Imbens (2016)

Causal trees: covariate adjustment

- ▶ This will perform well for estimating treatment effects if treatment is unconfounded within leaves
- ▶ As suggested in Powers et al. (2018), you can perform additional adjustment
 - ▶ Propensity score stratification within leaves
 - ▶ Other approaches to confounding adjustment certainly possible
 - ▶ Requires larger amount of data within leaves
- ▶ Can also incorporate propensity scores into choice of splits
 - ▶ Ensures individuals in same leaf have similar PS values
 - ▶ No longer maximizes heterogeneity of treatment effect

Causal forests: inference

- ▶ Inference in random forests models is typically very hard!
- ▶ Wager and Athey (2018) show how inference can be performed for random forests and causal forests
- ▶ Sample splitting is used such that
 1. Part of the data is used to find splits, i.e. tree structure
 2. Other part of the data is used to estimate treatment effects within leaves
- ▶ They show this leads to asymptotic normality of results with variance estimated by the infinitesimal jackknife (Wager et al. 2014)
- ▶ Implemented in the R package `grf`

Causal forests

- ▶ Throughout we've discussed creating splits for a single tree, but generally this is repeated a large number of times and results are averaged over all trees (as in random forests)
- ▶ We described the simplest type of causal forest
- ▶ Many extensions have been proposed that might perform better empirically
 - ▶ See Athey et al. (2019) for some ideas
- ▶ See also Powers et al. (2018) for other related algorithms such as boosting and MARS that are based on similar ideas

From causal trees to forests

- ▶ The most recent version of these causal forests involves combining causal forests with the R-learner
- ▶ Can create a pseudo-outcome as in the R-learner (also using regression trees to estimate $e(X)$ and $m(X)$)
- ▶ As in the R-learner, minimize

$$\tau(\cdot) = \underset{\tau}{\operatorname{argmin}} \left\{ \widehat{L}_n(\tau(\cdot)) + \Lambda_n(\tau(\cdot)) \right\}$$

- ▶ And now, the splits of the tree can be chosen to minimize this quantity

Tune a tree model for causal inference

- ▶ In ML models, a crucial step is to use cross-validation to tune hyperparameters: split the data into training (build model) and testing data (check model)
- ▶ In prediction problems, the standard performance metric is prediction MSE
- ▶ Similarly for an estimator of a causal estimand, say a CATE estimator $\hat{\tau}(x)$, we may use a MSE:

$$L(\hat{\tau}) = E[(Y_i(1) - Y_i(0) - \hat{\tau}(X_i))^2].$$

- ▶ But wait... this is usually not possible in causal inference problems, because even in the test data we do not know the **true causal effect** (Rolling and Yang, 2014; Athey and Imbens, 2016)
- ▶ So we would need approximations to the truth

Honesty Criterion

Athey and Imbens, 2016; Athey, Tibshirani, Wager 2018

- ▶ Honesty criterion: a sample can only be used to estimate τ or decide how to build the model (e.g. where to place the splits in trees), but not both.
- ▶ **Intuition: avoid using data twice**
- ▶ Implementation: the study sample is divided into three subsamples: two for training (one for building the tree and one for estimating causal effects) and one for testing
- ▶ Wager and Athey (2018) devised two tree-based *honest* procedure to estimate CATE: (i) double-sample (outcome) tree, and (ii) propensity tree (discuss later)
- ▶ Honesty is important to achieve asymptotic normality and unbiasedness.

Machine Learning and Causal Inference: Key Insights

- ▶ Machine learning greatly expands the toolbox for outcome modeling
- ▶ But machine learning **does not magically solve the fundamental problem of causal inference**
- ▶ The key issues in causal inference — overlap, balance, unconfoundedness — remain the same and requires more care
- ▶ To adapt machine learning methods to causal inference, one has to adapt to those key issues.
- ▶ Key insights:
 - ▶ Sample splitting: for building model and for estimating effects
 - ▶ Double learning: combine both PS model and outcome model for causal inference with high-dimensional data
 - ▶ Flexible (outcome) modeling

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